Pablo Alberto Duque Marín

AI Engineer

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Portfolio: pabloduque.dev — case studies, demos and labs on real data
Available: freelance now · full-time from Jan 2027 · remote

PROFILE

I take generative AI to production on AWS and measure statistically whether it works. I design document, RAG and agent pipelines on Amazon Bedrock, with Terraform infrastructure and automated tests. I measure whether they work with protocols fixed before looking at the data, and I publish negative results too.

EXPERIENCE

AI pipeline & infrastructure engineer (internship) Jun 2026 – present
Softart · master's internship: Jun–Jul (extracurricular) and Sep–Dec 2026 (curricular) *Remote, Colombia*

- Responsible for the backend and the AI pipeline that takes a traffic-insurance (SOAT) claim to an analysis ready for the lawyer a median of 28 s after the case is assigned (18 test runs), for a legal-sector client (Colombia). I also developed the React front end.
- Event-driven orchestration with EventBridge, DynamoDB and SQS, about 60 AWS resources in Terraform; gapless case numbering under 20 simultaneous registrations (DynamoDB transactions); captcha, write-only secrets and no personal data in logs.
- Policy branch with Amazon Textract and deterministic rules, with no LLM on purpose: extraction is not interpretation. Medical branch with Claude through Bedrock's Converse API and criteria versioned in Prompt Management; each case is summarised in 10 facts, each with its verbatim citation.
- A deterministic rule sends to human review any case whose medical record shows no clear link to the accident, keeping the model's original answer. Advised against displaying the model's "confidence": it clustered at 95, 85 and 0 with no defined scale.
- Backend tests from 642 to 845 in 2 weeks and coverage on new code from 38 to 84%, including synthetic medical records with an expected verdict against the deployed function; cold start from 867 to 545 ms.
- Accounting agent (under construction): architecture of an income-tax return MVP (Colombian forms F-110 and 2516) in which the LLM never calculates declarable figures: a deterministic Python engine with Decimal arithmetic, RAG only to explain, and mandatory human approval. Design of 27 stories, 20 decisions and 11 diagrams, and increment 0 done, with 79 tests and green CI.

Cloud infrastructure engineer Jul – Aug 2026
Freelance contract · tax SaaS (Spain) *Remote*

- Moved a hand-built RAG tax assistant from us-east-1 to Frankfurt for GDPR: migration complete and verified in Frankfurt within 9 days of the diagnosis (21–29 Jul); region-related defects after the cutover were fixed the same day.
- Reorganised the infrastructure as modular, multi-region Terraform (9 modules, remote state isolated per region and environment): 131 resources and a zero-diff plan against what is deployed.
- Replaced both end-of-life generative models with Claude Sonnet 4.5 and Haiku 4.5 on EU inference profiles, and moved the vector store from Aurora pgvector to S3 Vectors: data and models inside the EU.

Data analyst 2022 – 2026
Tubos y Perfiles de Occidente S.A.S. *Dosquebradas, Colombia*

- Built Python ETL processes joining operational and commercial data from several internal and external sources.
- Replaced manual reporting with scheduled Python scripts and wrote the SQL behind recurring analyses used for decision-making.

Founder and general manager 2019 – 2022
Narnia Producciones *Chile*

- Ran an event production company: budgeting, profitability analysis and supplier negotiation.

Second lieutenant, unit commander 2011 – 2018
Colombian National Police *Colombia*

- Commanded operational police units across Colombia.

PROJECTS

The same 9 case studies as on pabloduque.dev, in the same order; client work is anonymised.

From document to ruling: an AI pipeline for traffic-insurance claims case study · Private client code
Softart · legal-sector client · Colombia · 2026. **28 s** median from case assignment to an analysis ready for the lawyer (18 test runs). Policy branch with Textract and rules; medical branch with Claude on Bedrock, watched by a deterministic rule.

Moving a RAG tax assistant to Frankfurt for GDPR case study · Private client code

Freelance contract · tax SaaS · Spain · Jul–Aug 2026. **9 days** from diagnosis to a complete, verified migration in Frankfurt. Modular, multi-region Terraform, upgraded Bedrock agents and a move from Aurora pgvector to S3 Vectors.

Cotejo: checking cited case law before filing case study · Private repository

Own project · legal tech · Colombia · 2026. **0 / 500** false rejections across 500 real rulings. Checks every ruling cited in a brief against official sources: that it exists, was not annulled and is quoted verbatim.

Invoices to Excel: rules versus a local LLM case study · github.com/pabdus/facturas-a-excel · live demo · lab

Own project · documents · 2026. **240 / 240** exact fields with a local LLM and with Claude on Bedrock (rules: 67%). Colombian invoices to JSON and Excel with accounting checks, measured field by field. Live demo on AWS.

Fault detection in industrial processes Ongoing case study · github.com/pabdus/tfm-fdd-core · lab

Master's thesis · UNIR · 2026. **17 / 18** faults identical to Yin (2012) · verdict: not verified. PCA with T^2 and SPE on Tennessee Eastman, blind verification against the literature and comparisons with the protocol fixed in advance.

A critical evaluation of CTGAN case study · github.com/pabdus/ctgan-adult-critical-evaluation

Research · UCI Adult · 2026. **62%** impossible values, favourable metrics. Wasserstein and JSD looked good while 62% of the generated values were impossible.

SMOTE versus CTGAN under extreme imbalance case study · github.com/pabdus/proyecto-02-obesidad-brfss

Research · BRFSS 2023 · 2026. **≈ 0.02** macro-F1 difference between CTGAN and SMOTE: no real gain. 3 models × 4 strategies on 289,693 records, measured with macro-F1.

Urban–rural gaps in Colombia case study · github.com/pabdus/proyecto-01-desigualdad-colombia

Research · DANE 2018 census · 2026. **6.6×** rural versus urban illiteracy (aged 15 and over). Z-tests, χ^2 and logistic regression on census microdata: 15.4 M people in five departments.

Accounting agent: the model drafts, the engine calculates Under construction case study · Private repository

Internship · Softart · 2026. **79** passing tests in increment 0 (money engine and NIT). Income-tax return MVP (Colombian forms F-110 and 2516) from a trial balance, with deterministic calculation and human approval.

EDUCATION

MSc in Artificial Intelligence 2026 – Feb 2027
UNIR (Spain, online) · thesis on industrial fault detection *In progress · ends Feb 2027*

MSc in Applied Statistics 2025 – present
Universidad Cuauhtémoc (Mexico, online) *In progress*

BSc in Data Science 2023 – present
Universidad Compensar (Colombia, online) · in progress, 4th semester *In progress*

Data Science bootcamp (800+ h) 2022 – 2023
Henry *Completed*

Police Administration and systems technician 2007 – 2011
National Police · IES-CINOC *Completed*

SKILLS

Real level of use: production, own projects or coursework.

Languages: Python, SQL, HCL · Terraform *(production)*; R *(coursework)*

Generative AI: Amazon Bedrock, Claude Sonnet / Haiku 4.5, Prompt Management, Knowledge Bases · S3 Vectors *(production)*; Ollama · transformers *(projects)*

Documents: Amazon Textract *(production)*; pypdf · rapidfuzz, JSON Schema *(projects)*

ML and deep learning: scikit-learn, XGBoost, PyTorch · CTGAN *(projects)*; Keras · TensorFlow, spaCy *(coursework)*

Statistics: SciPy *(projects)*; statsmodels, R survey *(coursework)*

Data: pandas · NumPy, DynamoDB, PostgreSQL · MySQL *(production)*; Parquet *(projects)*

AWS cloud: Lambda, EventBridge · SQS · SES, S3 · CloudFront · Route 53, Cognito · AppSync · Amplify *(production)*

CI and quality: GitHub Actions, pytest · mutation testing, SonarCloud *(production)*

Observability: CloudWatch Logs, CloudTrail *(production)*

Model evaluation: F1-macro · AUC, TSTR · DCR · MIA, Wilcoxon · Holm · TOST *(projects)*

LANGUAGES

Native Spanish · technical English (B1–B2): I read and write documentation